EXEC 12T: detected-hit timing and local position reconstruction

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Abstract

The 4th-hit gate reproduces EXEC 11 exactly. The 20th detected hit yields a larger differential-time width but a steeper position slope, improving the leave-one-position-out X RMS68 from 13.34 to 10.32 mm. These are intrinsic simulation predictions.

1 Definitions and scope

The 4th and 20th detected hits are order statistics of accepted optical hits. They are not the 20% constant-fraction discriminator used by Lv et al. A detected-hit order statistic selects a row in the hit-time sequence; CFD acts on a reconstructed waveform crossing and Lv et al. combine 64 channels.

2 Provenance and reproduction gate

The fine scan contains 41 positions and 3000 events per position. Logs confirm EJ-204/OPSC-101, Top readout, zero added jitter, and data commit `f431c01`. The EXEC 11 4PE slope gate passes exactly: 6.8525923075 ps/mm in both products.

3 Fourth versus twentieth detected hit

metric	4th	20th
efficiency mean	1.000	1.000
slope ps/mm	5.880	9.673
mean delta-t RMS68 ps	78.470	99.837
mean CV X RMS68 mm	13.344	10.321
max abs bias mm	0.898	2.243
mean t+ RMS68 ps	47.992	65.677

At both thresholds every event passes, so native and matched comparisons are identical. The 20th hit broadens differential and common timing, but its position response is steeper. The resulting position resolution improves even though its raw time spread worsens.

4 Threshold sweep

Thresholds 1–30 all retain full efficiency in this high-light local scan. Increasing threshold generally steepens the calibration and improves X RMS68, while calibration nonlinearity rises strongly. No optimum is declared because electronics, SPTR, waveform formation, and experimental validation are absent.

5 Window-dip supplement

The EXEC 08b points are explanatory only and are not used for calibration. The nearest-channel mean count spans 349.4–413.2; the window-centred point retains the known local light deficit.

6 Comparison with Lv et al.

Lv et al. study an EJ-200 plate with 64 SiPMs, waveform-level 20% CFD, NPE weighting, 2016 geometric pairs, and a CNN. This work studies an EJ-204 bar, one adjacent Top pair, empirical one-dimensional calibration, and detected-hit order statistics. Numerical resolutions are not directly comparable.

7 Limitations and conclusion

This is a simulation prediction with no SPTR, electronics waveform, real CFD, noise, dark counts, cross-talk, afterpulsing, synchronization uncertainty, or experimental validation. The twentieth detected hit improves local X reconstruction in this dataset, but it does not improve the underlying raw timing width and it must not be described as 20% CFD.